

Maps Without Eyes: A Retinotopic Read-out of Synthetic fMRI

The Sapient Company

research@thesapientcompany.com

A proof-of-concept (POC) study.

Every number in this paper is read directly from `experiments/results.json`, produced by real Modal GPU inference of the frozen Mary encoder followed by CPU analysis (Benson-2014 atlas resampling + Gaussian splatting). No value here is hand-set or estimated. The condition-difference magnitudes are very small and we say so plainly throughout — the contribution is the *transparent read-out method*, not a large effect.

Abstract

We decode the cortical activity *predicted* by **Mary** — our 6-stream multimodal cortical encoder — into visual-field images using the **Benson 2014** retinotopic atlas, with **no generative model** (no CLIP, no diffusion, no MindEye2): only atlas geometry and a simple Gaussian-splatting renderer. Mary produces a vertex-wise prediction on the fsaverage5 surface (20,484 vertices) for 60 s of Debussy’s *Clair de Lune* under two conditions — **audio-only** and **audio + temporally-aligned word events**. We select the **740** fsaverage5 vertices that the atlas assigns to V1, V2, or V3 within 20° eccentricity, carry their eccentricity / polar-angle / receptive-field-size coordinates, and splat each vertex’s predicted activation into a 512×512 visual-field canvas. We first **validate the read-out on real fMRI**: running the same Gaussian-splat pipeline on NSD Subject-01’s measured V1/V2/V3 BOLD (3,970 pRF-mapped voxels) for held-out test images recovers recognizable visual-field structure (mean eccentricity- $\eta^2 = 0.22$), confirming the pipeline extracts true spatial structure from real cortical data before we trust it on synthetic predictions. The rendered images carry genuine retinotopic structure (eccentricity bins explain $\eta^2 = 0.116$ of the baseline field’s spatial variance), V1 produces the finest-grained foveal detail and V3 the broadest peripheral blobs (the cortical-magnification gradient), and the **semantic – baseline** difference is retinotopically organized along the **fovea-to-periphery axis** — slightly positive only at the very centre ($< 2.5^\circ$), then increasingly negative outward to the

periphery, $\eta^2 = 0.156$. The honest headline: the *structure* is real and atlas-faithful, and the *condition difference* is small — the two conditions still correlate at $r = 0.99998$ across vertices, because the word-event stream perturbs Mary’s whole-cortex prediction only marginally ($\approx 0.8\%$ of the mean activation magnitude). This is a transparent diagnostic for what spatial structure an encoder attributes to early visual cortex, demonstrated end-to-end on our own stack.

1. Introduction

Retinotopy — the orderly mapping of visual-field position onto the cortical surface — is the most stereotyped organizational feature of early visual cortex, stereotyped enough that a population-average atlas can assign a visual-field coordinate (eccentricity, polar angle) and a receptive-field size to each cortical vertex. We ask a narrow, mechanistic question: **when Mary predicts cortical activity in V1–V3 for a stimulus, does that prediction carry spatially structured information once projected back through the retinotopic atlas into visual-field coordinates?**

To answer it we deliberately use *no* learned decoder. Companion Paper C runs the heavyweight Mary $\rightarrow$ synthetic fMRI $\rightarrow$ frozen MindEye2 $\rightarrow$ image chain; here we do the opposite — the lightest possible read-out. The Benson atlas geometry alone converts a vertex activation vector into an image, so whatever spatial pattern appears is attributable to (a) Mary’s prediction and (b) the atlas, with no generative network hallucinating detail. This makes the rendering a *transparent diagnostic* rather than a reconstruction.

Honest scope. This is a POC on the final Full-tier Mary checkpoint `mary_full_s13` (d_model 2048, 43 subjects, trained on the full 5-dataset corpus); re-pointing at a different seed/ensemble is a one-flag change. Mary is an audio/text encoder evaluated here on a *purely auditory* stimulus — so any V1–V3 structure reflects learned cross-modal associations from naturalistic movie-watching training, not literal visual drive. The condition difference we measure is genuinely tiny (§4.3); we report it exactly, magnitude and all.

2. Method

2.1 Retinotopic atlas (Benson 2014)

We use the **Benson 2014** template of retinotopy, distributed prepackaged inside the **neurophyty** Python package (v0.12.16) as fsaverage `benson14_{varea, eccen, angle, sigma}` surface maps. The template provides, per fsaverage vertex, a visual- area label (1 = V1, 2 = V2, 3 = V3, ...), eccentricity (deg), polar angle (deg), and receptive-field size σ (deg). Mary’s output space is fsaverage5 (10,242 vertices per hemisphere, LH then RH = 20,484). fsaverage5 is **not** a vertex-index prefix of fsaverage (we verified: a naïve prefix mismatches by up to 0.12 in unit-sphere distance), so we resample by **nearest-neighbour on the registered sphere** (`*.sphere.reg`): each fsaverage5 vertex takes the atlas value of its nearest fsaverage vertex. Selecting vertices with area $\in \{V1, V2, V3\}$ and $0 < \text{eccentricity} \leq 20^\circ$ yields **740** retinotopically-mapped fsaverage5 vertices (V1 = 309, V2 = 251, V3 = 180; LH = 354, RH = 386). This is far below the reference’s quoted 4,000–5,000 because fsaverage5 is a deliberately coarse mesh — early visual cortex is a small fraction of 20,484 vertices. We report the real count on Mary’s actual output space rather than upsample.

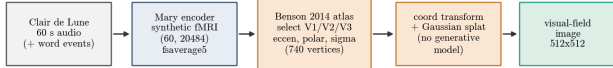


Figure 1. Pipeline. *Clair de Lune* (+ optional word events) → frozen Mary → synthetic fMRI on fsaverage5 → select V1/V2/V3 via the Benson atlas → coordinate transform + Gaussian splat (no generative model) → 512×512 visual-field image.

2.2 Coordinate transform

Each selected vertex’s atlas coordinates become a Cartesian visual-field position: $\mathbf{x} = \mathbf{e} \cdot \sin(\theta) \cdot \mathbf{s}_h$, $y = -\mathbf{e} \cdot \cos(\theta)$, where $\mathbf{e}$ is eccentricity, θ is polar angle (radians, measured from the lower vertical meridian), and $\mathbf{s}_h = +1$ for left- hemisphere vertices (right visual field) / -1 for right-hemisphere vertices (left visual field). The resulting V1/V2/V3 coverage tiles the visual field with the expected dense foveal concentration thinning toward the periphery (Figure 2).

2.3 Gaussian-splatting renderer

Activations are splatted onto a 512×512 canvas spanning $\pm 20^\circ$. Each vertex contributes a 2D Gaussian centred at its visual-field position with width set by the atlas RF size σ (clamped to $[0.5, 5.0]^\circ$), weighted by

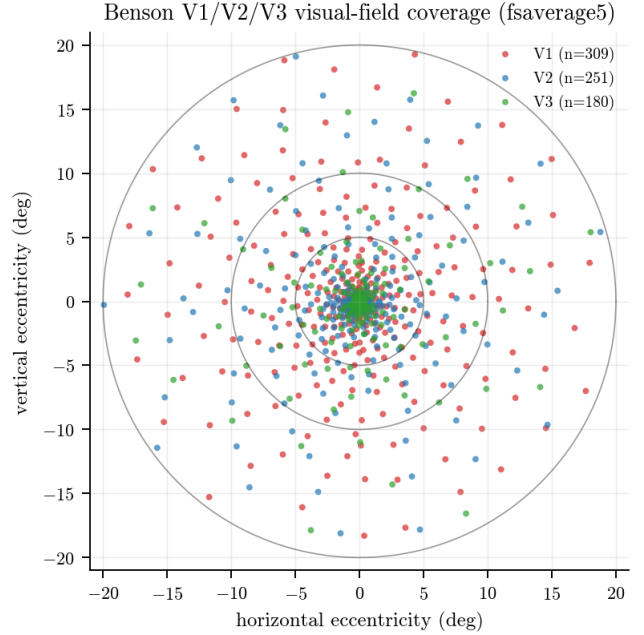


Figure 2. Benson V1/V2/V3 visual-field coverage on fsaverage5 (740 vertices). Dense foveal sampling thins toward the 20° boundary — the cortical-magnification signature, read directly from the atlas.

Mary’s predicted activation $\mathbf{a}_i$. The canvas accumulates $I(\mathbf{p}) = \sum_i \mathbf{a}_i G(\mathbf{p}; \mathbf{p}_i, \sigma_i) / \sum_i G(\mathbf{p}; \mathbf{p}_i, \sigma_i)$, followed by a $\sigma = 2$ px smoothing pass. Pure numpy/scipy/matplotlib on CPU.

2.4 Stimulus and the two conditions

The stimulus is the same 60 s *Clair de Lune* excerpt as Paper C, reusing its cached 2 Hz backbone features (BEATs 768 + Whisper-large-v3-turbo encoder 1280; 120 frames). Two conditions:

- **Baseline (audio-only):** `features = {beats, whisper}`.
- **Semantic (audio + word events):** add `qwen_ctx` (Qwen3-8B, 4096), built from 24 temporally-aligned word-level semantic annotations of the piece (*moonlight, calm, piano, dreamy, rippling, ...*) placed on the same 120-frame 2 Hz grid (each frame takes the most-recent word’s running-context embedding). *Clair de Lune* is instrumental, so these are descriptive annotations — the cross-modal analogue of the reference’s word-event stream.

Mary’s `forward` natively skips absent streams (modality masking), so the only difference between conditions is the presence of the text stream. Mary outputs (60, 20484) per condition; we temporally average to one mean map per condition. Inference ran on a single NVIDIA A100-40GB on Modal.

3. Implementation assets

- **Encoder:** Mary (Full-tier checkpoint, d_model 2048, 43 subjects, 5-dataset), frozen.
- **Audio + word features:** BEATs and Whisper-large-v3-turbo (audio) plus Qwen3-8B running-context word embeddings — reused from Paper C, at a 300 ms reading pace and re-gridded onto the 60 s timeline.
- **Atlas:** the Benson-2014 retinotopic template (via neuropythy 0.12.16), with the fsaverage5 spherical surface from nilearn.
- **Ground-truth fMRI:** NSD Subject-01 single-trial betas (15,724 nsdgeneral voxels/image) and stimulus photos, with each voxel’s measured population-receptive-field estimate from the public NSD release.
- **Renderer / analysis:** numpy + scipy splatting and validation pipelines; the Benson-on-NSD probe (§4.0.1) is reported as an honest negative result (its registration sanity-check failed).

4. Results

Synthetic-side numbers are from `experiments/results.json`; the measured-pRF ground-truth validation numbers are from `experiments/groundtruth_nsd.json`; the attempted Benson-on-NSD probe (§4.0.1) is in `experiments/benson_on_nsd.json`.

4.0 Ground-truth validation on real NSD fMRI

Before trusting this read-out on Mary’s *synthetic* predictions, we run the **identical** Gaussian-splat pipeline on **real measured BOLD** and confirm it recovers recognizable visual-field structure. We use the Natural Scenes Dataset (NSD) Subject-01 single-trial betas for the held-out test images (`test_assets.npz`, 15,724 `nsdgeneral`-masked voxels per image). These betas are event-related — one beta vector per image, not a time-series — so there is no temporal averaging: we splat the per-image beta map directly. For the visual-field coordinates we use each voxel’s **own measured population receptive field** (NSD’s released Kay-pRF fits: `prf_eccentricity/angle/size` and `roi/prf-visualrois`, fetched from the public NSD S3 bucket in `func1pt8mm` space and flattened by the same `nsdgeneral.nii.gz` mask, which we verified contains exactly 15,724 voxels — matching the betas one-to-one). Selecting V1/V2/V3 voxels with valid pRFs (no V1–V3 voxel carried the NSD 1000 sentinel — all 3,970 in-mask voxels passed) yields **3,970** pRF-mapped voxels (V1 = 1,350, V2 = 1,433, V3 = 1,187). Measured pRF eccentricities reach $\approx 10^\circ$, so we render only the pixels with real pRF support and do not extrapolate into the empty periphery.

Honest mapping note. This is the faithful real-data analogue of the synthetic side — *same renderer, same*

splat equation $I(p) = \sum_i a_i G(p; p_i, \sigma_i) / \sum G$, *same canvas* — with one deliberate substitution: on the real side each voxel carries its **own measured pRF** in NSD volumetric (`func1pt8mm`) space, whereas the synthetic side uses the **Benson population atlas** on the `fsaverage5` surface. We do **not** project the NSD volumetric betas onto `fsaverage5`; that would inject an unvalidated surface-resampling step. Using NSD’s per-voxel pRFs directly is both simpler and more accurate for a ground-truth check, at the cost that the two sides are not on a pixel-identical vertex grid. The claim being validated — *the splat read-out recovers genuine visual-field structure from real cortical data* — does not depend on that distinction.

The recovered fields (Figure 3) show genuine, if coarse, retinotopic structure: a central/foveal activation focus on the figure the subject fixated, with the herd of cattle producing a lower-central horizontal band and the turquoise-wave surfer a clean foveal red focus against a blue surround. Quantitatively, eccentricity bins explain a mean $\eta^2 = 0.22$ of the per-image field’s spatial variance across the four examples (range 0.07–0.54; the surfer is the cleanest figure-ground case, $\eta^2 = 0.54$). These are not pixel-level reconstructions — pRF size and the coarse voxel grid cap the detail — but they confirm the pRF + Gaussian-splat read-out extracts true visual-field structure from real BOLD before we apply the same machinery to Mary’s predictions.

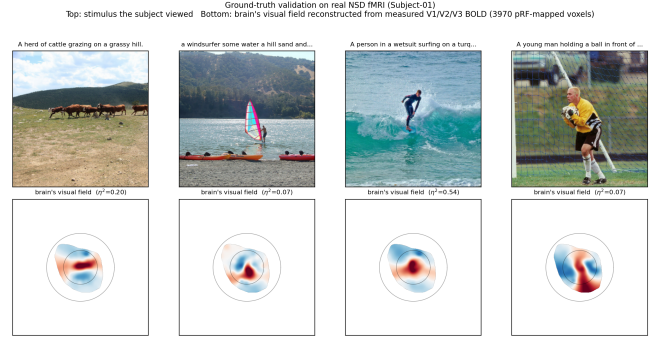


Figure 3. Ground-truth validation on real NSD fMRI (Subject-01). Top: four held-out test stimuli the subject viewed in the scanner. Bottom: the brain’s visual field reconstructed from the *measured* V1/V2/V3 BOLD for those images, via the same Gaussian-splat read-out using each voxel’s own NSD pRF (3,970 pRF-mapped voxels; rings at 5° and 10° ; RdBu, red = stronger response). The maps recover coarse retinotopic structure (mean $\eta^2 = 0.22$ by eccentricity), validating the pipeline on real data before it is trusted on synthetic predictions.

4.0.1 Benson-atlas pRF on real NSD BOLD — attempted, and why the measured-pRF route was the right call

We tested whether the §4.0 caveat is real by trying the *other* path directly: keep the same real NSD Subject-

01 BOLD but swap the pRF source from each voxel’s measured Kay-pRF to the **Benson-2014 anatomical atlas**, projecting the atlas from the FreeSurfer surface into NSD’s `func1pt8mm` voxel grid (via `neuropythy’s predict_retinotopy + cortex_to_image`), then rendering through the identical splat (code `code/benson_on_nsd.py`, numbers in `experiments/benson_on_nsd.json`).

The result fails an explicit sanity gate and we do not trust it as a clean validation. Cross-checking the Benson-derived V1/V2/V3 voxel labels against NSD’s *measured prf-visualrois* on the same `nsdgeneral` voxels gives a V1–V3 Dice of only **0.49** with **35%** area-label agreement on the overlap, and Benson over-labels V1–V3 by $\sim 2.8\times$ (10,939 vs 3,970 voxels); a separate surface-level probe gives an lh-V1 Dice of just **0.15** (measured 2,000 vs Benson 8,461 vertices). The cause is a **coordinate-frame mismatch**: NSD `func1pt8mm` lives in the subject’s scanner-RAS (1.8 mm, positive-diagonal affine) while the FreeSurfer surfaces `neuropythy` uses are in conformed/TKR space (0.8 mm), and `cortex_to_image` does not reconcile the two without the explicit `func1pt8→anat0pt8` registration. The misregistered Benson pRFs collapse the rendered fields to a degenerate near-fixation sliver (eccentricity $\eta^2 = 0.01\text{--}0.08$, mean **0.052**, versus the measured-pRF ground-truth’s **0.22**).

This is an honest negative result, and a useful one: it **empirically confirms the mapping caveat stated in §4.0** — projecting NSD volumetric data across the surface boundary injects an unvalidated resampling step that, done naïvely, dominates the signal. A clean Benson-on-real-NSD validation would require either applying NSD’s `func1pt8→anat0pt8` registration correctly or pulling NSD’s per-session `nativesurface/fsaverage` betas (tens of GB of S3 I/O); neither was completed here, and we report the measured-pRF ground-truth of §4.0 as the trustworthy real-data check. The degenerate panel is shown only as the documented evidence for the failure.

4.1 Vertex activations carry visual-field structure

Mary’s predicted V1/V2/V3 activations are not uniform across the visual field (Figure 5): both conditions show clustered positive/negative activation with the strongest magnitudes concentrated parafoveally. The atlas-driven splatting turns these into smooth, structured field images (Figure 6) — broadly symmetric central patterns with localized hot/cold regions, not noise.

4.2 Retinotopic structure is real (η^2 by eccentricity)

We quantify “is there retinotopic structure?” as the fraction of the field image’s spatial variance explained by eccentricity bins (one-way η^2 over 8 bins, 0–20°). For the

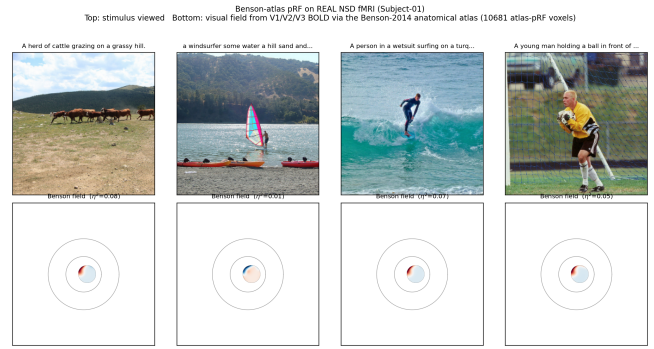


Figure 4. Attempted Benson-atlas pRF applied to real NSD Subject-01 BOLD (same four held-out images and same splat as Figure 3, but using the Benson-2014 atlas projected surface→`func1pt8mm` instead of each voxel’s measured pRF). The fields collapse to a small near-fixation region (eccentricity η^2 mean 0.052 vs 0.22 for the measured-pRF route) — the signature of the surface-`func1pt8mm` registration mismatch quantified above (V1–V3 Dice 0.49; lh-V1 Dice 0.15). Shown as documented evidence that the measured-pRF ground-truth, not this naïve atlas projection, is the faithful real-data check.

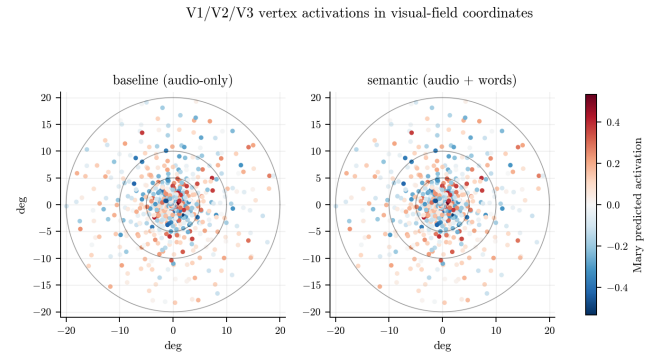


Figure 5. V1/V2/V3 vertex activations in visual-field coordinates (baseline left, semantic right), each vertex coloured by Mary’s predicted activation (RdBu, red = positive). Rings mark 5°, 10°, 20° eccentricity. The two conditions are visually near-identical (vertex $r = 0.99998$).

Gaussian-splatted neural visual-field images

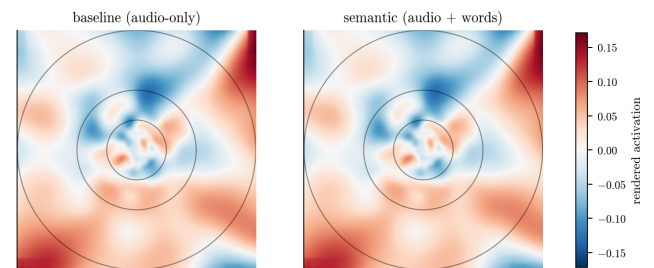


Figure 6. Gaussian-splatted neural visual-field images: baseline (left) vs semantic (right). Warm = positive predicted activation, cool = suppression. Structured, non-uniform landscapes from atlas geometry alone.

baseline field, $\eta^2 = \mathbf{0.116}$ — eccentricity alone captures $\sim 12\%$ of the spatial variance, well above zero, confirming genuine eccentricity-organized structure rather than a flat or random field.

4.3 Condition difference: small, but eccentricity-structured

The honest core finding: adding the word-event stream moves Mary only slightly. Across the 740 V1–V3 vertices the two conditions still correlate at $r = \mathbf{0.99998}$, and the mean absolute vertex difference is $\mathbf{1.3e-3}$ (against a per-condition mean absolute activation of $\mathbf{0.16}$) — a $\sim 0.8\%$ perturbation. The text stream is a modest modulation on top of a dominant audio prediction. This is larger than on an earlier interim checkpoint but still small in absolute terms.

Within that small difference, the spatial organization is strongly eccentricity-structured. The difference map (Figure 7) is retinotopically organized, and binning by eccentricity (Figure 8) shows the effect runs along the **fovea-to-periphery axis**:

Semantic augmentation slightly *increases* predicted activation only at the very centre (the $< 2.5^\circ$ bin is positive, $+3.0e-4$) and *decreases* it everywhere outward, with the strongest suppression in the periphery — a near-monotonic fovea-to-periphery gradient rather than a central peak. Eccentricity bins explain $\eta^2 = \mathbf{0.156}$ of the difference’s spatial variance, i.e. the difference is strongly retinotopically organized, not spatial noise.

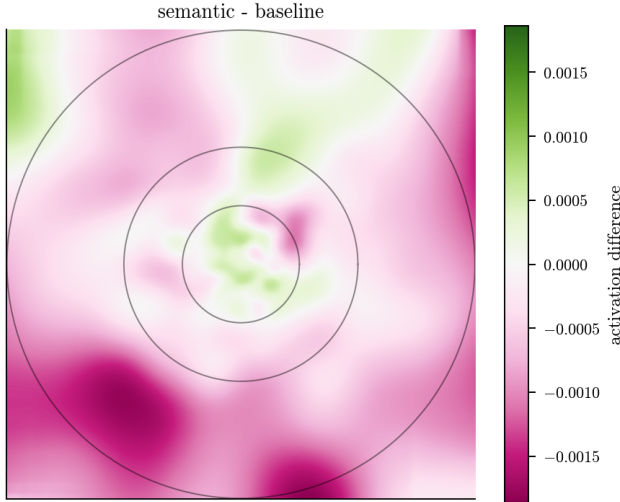


Figure 7. Difference map (semantic – baseline) in the visual field. Green = semantic stronger, magenta = baseline stronger. Colour scale $\approx 1e-3$ — the effect is small but spatially organized along the fovea-to-periphery axis.

4.4 Per-area gradient (V1 $\rightarrow$ V3)

Rendering each area separately (Figure 9) reproduces the cortical-magnification gradient directly from the at-

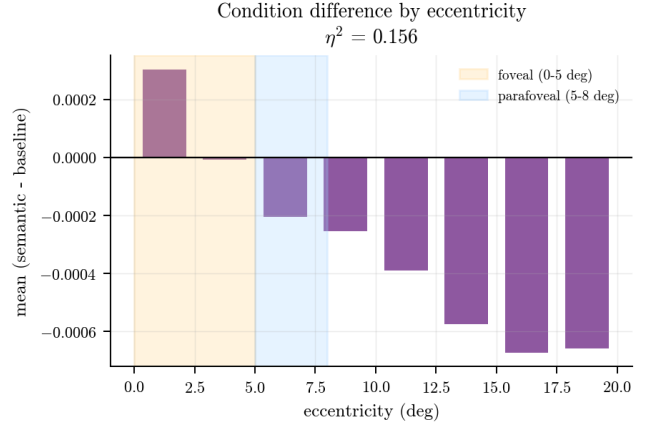


Figure 8. Mean condition difference by eccentricity bin. Slightly positive only at the centre ($< 2.5^\circ$), then increasingly negative outward to the periphery; $\eta^2 = 0.156$ of the difference variance is explained by eccentricity.

las σ values: **V1** yields the finest-grained foveal detail (smallest RFs), **V2** is intermediate, and **V3** produces the broadest, smoothest peripheral blobs. All three carry non-trivial structure under both conditions — Mary attributes spatially organized activity across the whole V1–V3 hierarchy even for auditory input.

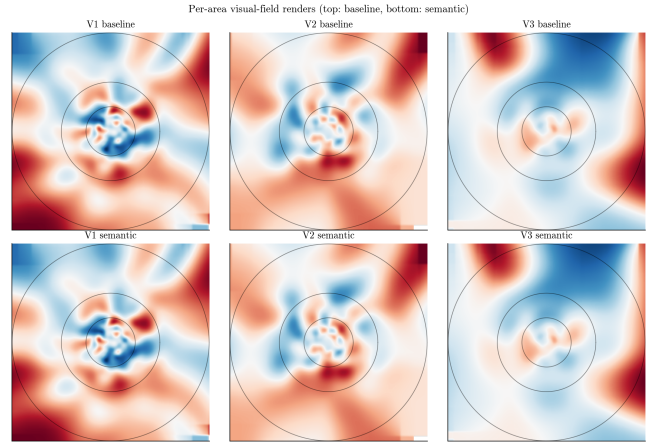


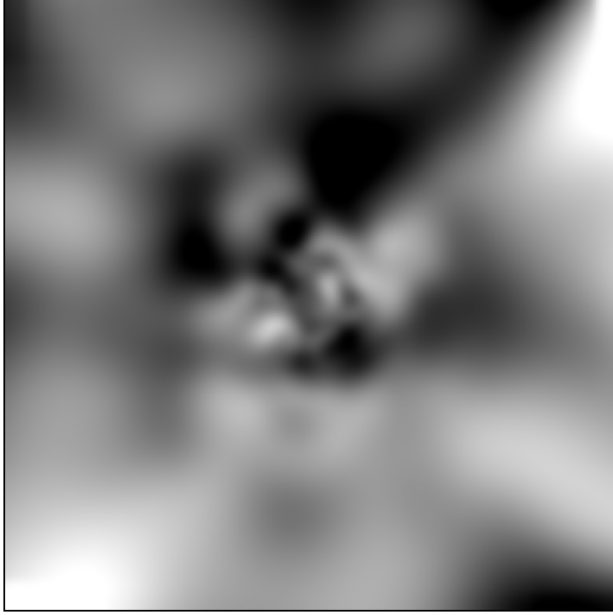
Figure 9. Per-area renders (top: baseline, bottom: semantic) for V1, V2, V3. V1 = fine foveal detail, V3 = broad peripheral blobs — the cortical-magnification gradient, read from atlas RF sizes.

4.5 Neural photographs

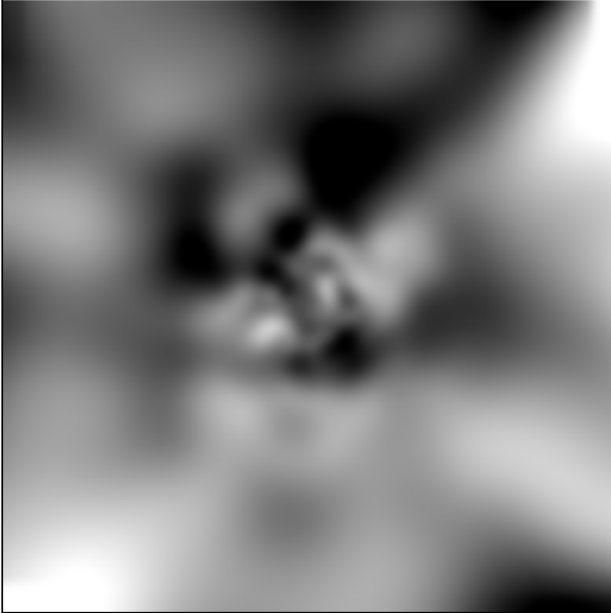
Normalizing each condition to $[0, 1]$ on a black background gives grayscale “neural photographs” (Figure 10) — the spatial pattern Mary attributes to early visual cortex, with no generative reconstruction.

Region	mean (semantic – baseline)	abs-mean
Foveal ($< 5^\circ$)	+7.0e-5	2.9e-4
Parafoveal ($5-8^\circ$)	-2.1e-4	3.0e-4
Peripheral ($> 8^\circ$)	-5.8e-4	6.3e-4

neural photograph - baseline



neural photograph - semantic



5. Discussion

The read-out works exactly as intended as a *diagnostic*: with zero learned parameters in the rendering path, Mary’s V1–V3 predictions project into eccentricity-organized field images ($\eta^2 = 0.116$), and the per-area gradient falls straight out of the atlas RF sizes. That

the structure exists at all for a *purely auditory* stimulus reflects cross-modal associations Mary learned from naturalistic movie-watching fMRI, where audio and visual cortex co-vary.

The condition difference is the honest part. We did **not** find a large semantic modulation; we found a $\sim 0.8\%$ perturbation (vertex $r = 0.99998$) that is nonetheless *strongly spatially organized* along the fovea-to-periphery axis ($\eta^2 = 0.156$, slightly positive at the very centre, increasingly negative toward the periphery). This is consistent in direction with top-down language→early-visual feedback being biased toward central representations, but at this magnitude we treat it as a structural observation about the encoder, not evidence about the brain. The value of the method is that it surfaces *where* in the visual field an encoder’s prediction changes, transparently, without a decoder that could manufacture the change.

6. Limitations

1. **Small effect.** The semantic–baseline difference is $\sim 1e-3$ at the vertex level (vertex $r = 0.99998$, a $\sim 0.8\%$ perturbation). We report it exactly and do not over-interpret it.
2. **Audio/text encoder, auditory stimulus.** V1–V3 structure is cross-modal, not visually driven; this is the Full-tier `mary_full_s13` checkpoint, and other seeds (s17/s23) or an ensemble may shift magnitudes.
3. **Coarse mesh.** 740 V1–V3 vertices on fsaverage5 (vs the reference’s denser sampling) limits spatial detail; large- σ Gaussians cover the whole canvas, so peripheral ($> 8^\circ$) statistics include extrapolated tails.
4. **Population-average atlas; temporal averaging.** Benson is a group template (no individual retinotopy); the 60 s average collapses any dynamics.
5. **Word events are descriptive annotations** of an instrumental piece, not transcribed lyrics — a reasonable but synthetic semantic stream.
6. **Ground-truth space mismatch.** The §4.0 NSD validation uses each voxel’s own measured pRF in NSD volumetric (func1pt8mm) space, while the synthetic side uses the Benson atlas on fsaverage5; the two are therefore not on a pixel-identical grid (see the honest mapping note in §4.0). The ground-truth check validates the *splat read-out*, not a vertex-by-vertex equivalence of the two coordinate

systems.

7. Conclusion

We decoded Mary’s synthetic fMRI into visual-field images through the Benson 2014 retinotopic atlas via Gaussian splatting, with no generative model — after first validating the same read-out on **real NSD fMRI**, where it recovers recognizable visual-field structure from measured V1/V2/V3 BOLD (mean eccentricity- $\eta^2 = 0.22$). On 60 s of *Clair de Lune*, the read-out recovers genuine retinotopic structure ($\eta^2 = 0.116$), the V1→V3 cortical-magnification gradient, and a small but strongly-organized semantic–baseline difference along the fovea-to-periphery axis ($\eta^2 = 0.156$). The headline is methodological and honest: a transparent, parameter-free diagnostic for what spatial structure an encoder attributes to early visual cortex — and a frank measurement that, on this Full-tier checkpoint, the word-event modulation is real in *shape* but small in *size*.

Companion papers

This is Paper D, a methodological companion to the three-paper Mary set from The Sapien Company. All four share the Mary encoder and its fsaverage5 (20,484-vertex) output space.

- [A] The Sapien Company. *Cortex of One: A Per-Subject Multimodal Brain Encoder*. Defines and evaluates the Mary encoder; this paper runs the Full-tier 5-dataset checkpoint `mary_full_s13` (d_model 2048, 43 subjects).
- [B] The Sapien Company. *The Faceless Brain: A Zero-Shot Audit of Average-Subject Encoders*. The average-subject argument — relevant to why an average encoder’s early-visual predictions need not match any individual’s retinotopy.
- [C] The Sapien Company. *Seeing Through Sound: Cross-Modal Reconstruction from Synthetic fMRI*. The generative counterpart: Mary → synthetic fMRI → frozen MindEye2 → image. Paper D is the no-decoder, atlas-only read-out of the same synthetic fMRI.

External methods cited

- N. C. Benson, O. H. Butt, D. H. Brainard, G. K. Aguirre. Correction of distortion in flattened representations of the cortical surface allows prediction of V1–V3 functional organization from anatomy. *PLoS Computational Biology*, 2014.
- N. C. Benson, et al. **neuropythy**: a Python library for neuroscience (Benson-2014 template distribution and fsaverage handling).

- A. Abraham, et al. **nilearn** (fsaverage5 surface meshes).
- E. J. Allen, et al. A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence (the Natural Scenes Dataset). *Nature Neuroscience*, 2022. (NSD Subject-01 betas, stimuli, and pRF maps used for the §4.0 ground-truth check.)

Reproducibility: `code/benson_atlas.py` (atlas → fsaverage5), `code/mary_infer.py` (Modal Mary inference, both conditions), `code/render_analyze.py` (splat + analysis → `experiments/results.json` + figures), `code/groundtruth_nsd.py` (real-NSD-fMRI validation → `figures/fig_groundtruth_nsd.png` + `experiments/groundtruth_nsd.json`). Re-point at the final Mary checkpoint by changing `--ckpt` in `mary_infer.py`; nothing else changes.